WORKED SOLUTION

STEP I 2004, Question 12

Probability and Statistics • Microships

STEP I 2004, Question 12

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/04-s1-q12

The key idea

The sample from group 1 tells us, approximately, the conditional probability that a chip came from machine A *given* that it is in group 1. So we write that conditional probability in terms of λ, p and q using Bayes' theorem, set it equal to the observed proportion, and solve for λ. The faulty sorting changes only the probability that a chip from each machine lands in group 1; the method is identical.

Solution

Pick a chip at random from the factory's output. Let A be the event that it was made by machine A (so $\mathrm{P}(A) = \lambda$ and $\mathrm{P}(A') = 1 - \lambda$), let P be the event that it is perfect, and let G be the event that it is placed in group 1. We are told $\mathrm{P}(P \mid A) = p$ and $\mathrm{P}(P \mid A') = q$.

A chip drawn at random from group 1 is a chip drawn at random from the output, conditioned on G. Since the sample is large, the observed proportion $\frac{2}{5}$ is a good estimate of $\mathrm{P}(A \mid G)$, and our estimate of λ is the value that makes $\mathrm{P}(A \mid G)$ exactly $\frac{2}{5}$.

Perfect sorting

Here G is the same event as P. By Bayes' theorem,

$$\mathrm{P}(A \mid P) = \frac{\mathrm{P}(P \mid A)\,\mathrm{P}(A)}{\mathrm{P}(P \mid A)\,\mathrm{P}(A) + \mathrm{P}(P \mid A')\,\mathrm{P}(A')} = \frac{\lambda p}{\lambda p + (1-\lambda)q}.$$

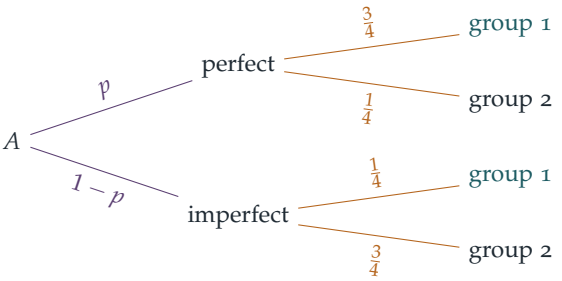
Setting this equal to $\frac{2}{5}$ and cross-multiplying (the denominator is positive):

$$\begin{aligned} 5\lambda p &= 2\lambda p + 2(1-\lambda)q \\ 3\lambda p + 2\lambda q &= 2q \\ \lambda &= \frac{2q}{3p+2q}, \end{aligned}$$

which is the required estimate, since $3p + 2q > 0$.

Faulty sorting

Now a chip reaches group 1 either by being perfect and correctly sorted, or by being imperfect and wrongly sorted. We assume the sorting errors happen independently of which machine made the chip, so that, given whether a chip is perfect, its sorting does not depend on its machine. The tree below shows the branches for a chip from machine A; machine B is the same with q in place of p.



Adding the two group-1 branches,

$$\mathrm{P}(G \mid A) = \tfrac{3}{4}p + \tfrac{1}{4}(1-p) = \frac{1+2p}{4}, \qquad \mathrm{P}(G \mid A') = \tfrac{3}{4}q + \tfrac{1}{4}(1-q) = \frac{1+2q}{4}.$$

So, by Bayes' theorem again (the factors of $\frac{1}{4}$ cancel),

$$\mathrm{P}(A \mid G) = \frac{\lambda(1+2p)}{\lambda(1+2p) + (1-\lambda)(1+2q)}.$$

Ask yourself Has the evidence changed? Should the $\frac{2}{5}$ still be used?

Yes: the sample was genuinely drawn from group 1, and $\frac{2}{5}$ of it genuinely came from machine A. What has changed is only our model of how chips got into group 1. So we set the new expression for $\mathrm{P}(A \mid G)$ equal to $\frac{2}{5}$:

$$\begin{aligned} 5\lambda(1+2p) &= 2\lambda(1+2p) + 2(1-\lambda)(1+2q) \\ \lambda\big[3(1+2p) + 2(1+2q)\big] &= 2(1+2q) \\ \lambda &= \frac{2(1+2q)}{5+6p+4q}. \end{aligned}$$

This is the new estimate of λ. Notice it is the first answer with p and q replaced by $\frac{1+2p}{4}$ and $\frac{1+2q}{4}$, the new probabilities of reaching group 1.

Checks

- If $p = q$ the machines are indistinguishable by quality, so group 1 should mirror production and λ should be $\frac{2}{5}$. Both formulae give this: $\frac{2p}{5p} = \frac{2}{5}$ and $\frac{2(1+2p)}{5+10p} = \frac{2}{5}$.

- Both estimates lie strictly between 0 and 1, since in each the numerator is positive and smaller than the denominator.
- A simulation of 4×10^5 chips with $\lambda = 0.3$, $p = 0.6$, $q = 0.8$ and faulty sorting gave 0.2667 of group 1 from machine A, against the formula's 0.2661; inverting the formula on the simulated proportion recovered $\lambda \approx 0.301$.

Common slips

- Equating $\frac{2}{5}$ to $\mathrm{P}(G \mid A)$ or to λp rather than to the conditional probability $\mathrm{P}(A \mid G)$.
- In the second part, forgetting the imperfect chips that are wrongly put into group 1, i.e. using $\frac{3}{4}p$ alone.

Beyond STEP

Choosing the parameter value that makes the observed data "fit exactly" is the *method of moments*; here it coincides with maximum likelihood estimation, which you will meet in a first statistics course. The faulty sorter is a simple model of measurement error (misclassification), a serious practical issue in medical testing, where false positives and false negatives distort the observed proportions in just this way. For more on Bayes' theorem and conditioning, see Blitzstein and Hwang, *Introduction to Probability*.

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